Questions

3.3 Find the total number of free electrons in a circular wafer of single crystal silicon that is 300 μm thick and 15 cm in diameter at $T = 300$ K.

3.6 In a boron-doped silicon layer with impurity concentration of $10^{17}/\text{cm}^3$, find the hole and electron concentrations at 27°C and 125°C. Is the result n- or p-type silicon?

3.11 Holes are being steadily injected into a region of n-type silicon (connected to other devices, the details of which are not important for this question). In the steady state, the excess-hole concentration profile shown in Fig. P3.11 is established in the n-type silicon region at room temperature. Here "excess" means over and above the thermal-equilibrium concentration (in the absence of hole injection), denoted p_{n0}. If $N_D = 10^{16}/\text{cm}^3$, $n_i = 1.5 \times 10^{10}/\text{cm}^3$, $D_p = 12\ \text{cm}^2/\text{s}$, and $W = 50$ nm, find the density of the current that will flow in the x direction.

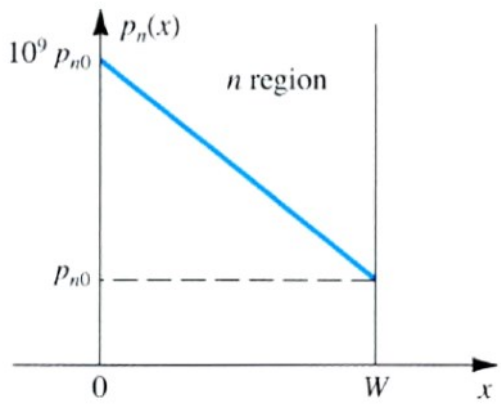


Figure P3.11

3.13 Calculate the built-in voltage of a junction in which the p and n regions are doped equally with 5×10^{16} atoms/cm^3. Assume $n_i = 1.5 \times 10^{10}/\text{cm}^3$. With the terminals left open, what is the width of the depletion region, and how far does it extend into the p and n regions? If the cross-sectional area of the junction is 2 μm^2, find the magnitude of the charge stored on either side of the junction.

3.16 In a pn junction for which $N_A \gg N_D$, and the depletion layer exists mostly on the lightly doped side with $W = 0.2$ μm, find V_0 if $N_D = 10^{16}/\text{cm}^3$. Also calculate Q_j for the case $A = 10\ \mu\text{m}^2$.

3.18 In a pn junction with $N_A \gg N_D \gg n_i$, sketch a plot of charge density on both sides of the junction. How does the sketch change if N_D is increased by a factor of 4?

3.19 Given that the electric field E has a triangular profile as shown in Fig. P3.19, and that the built-in potential, V_0, is given by the total area under the triangle, find an expression for the maximum electric field magnitude, E_{max}, in terms of the dopant concentrations and fundamental constants.

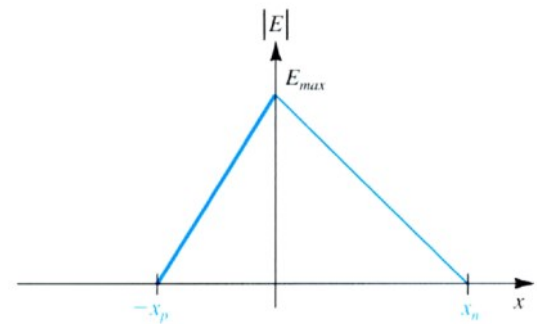


Figure P3.19

3.25 A p^+n junction is one in which the doping concentration in the p region is much greater than that in the n region. In such a junction, the forward current is mostly due to hole injection across the junction. Show that

$$I \simeq I_p = Aqn_i^2 \frac{D_p}{L_p N_D}\left(e^{V/V_T} - 1\right)$$

For the specific case in which $N_D = 10^{16}/\text{cm}^3$, $D_p = 10\ \text{cm}^2/\text{s}$, $L_p = 10$ μm, and $A = 10^4$ μm^2, find I_S and the voltage V obtained when $I = 0.2$ mA. Assume operation at 300 K where $n_i = 1.5 \times 10^{10}/\text{cm}^3$.

3.29 The junction capacitance C_j can be thought of as that of a parallel-plate capacitor and thus given by

$$C_j = \frac{\epsilon A}{W}$$

Show that this approach leads to a formula identical to that obtained by combining Eqs. (3.42) and (3.44) [or equivalently, by combining Eqs. (3.44) and (3.45)].

3.30 A pn junction operating in the forward-bias region with a current I of 0.4 mA is found to have a diffusion capacitance of 1 pF. What diffusion capacitance do you expect this junction to have at $I = 0.1$ mA? What is the mean transit time for this junction?